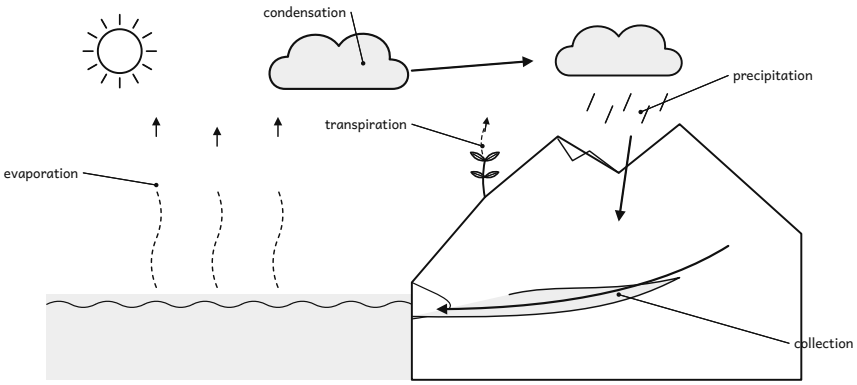
EARTH AND SPACE

Water for Life and the Water Cycle

Changes of state, the water cycle, fresh water in Bangladesh and desalination



In this pack

- **Learn it:** Water for life and the water cycle ★☆☆
- **Fact check:** The water cycle ★☆☆
- **Think it through:** Explaining the cycle ★★☆
- **Investigate:** Make a solar still ★★☆
- **Real-world case study:** Too much water, not enough to drink ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · B1 · WATER CYCLE

Water for life and the water cycle

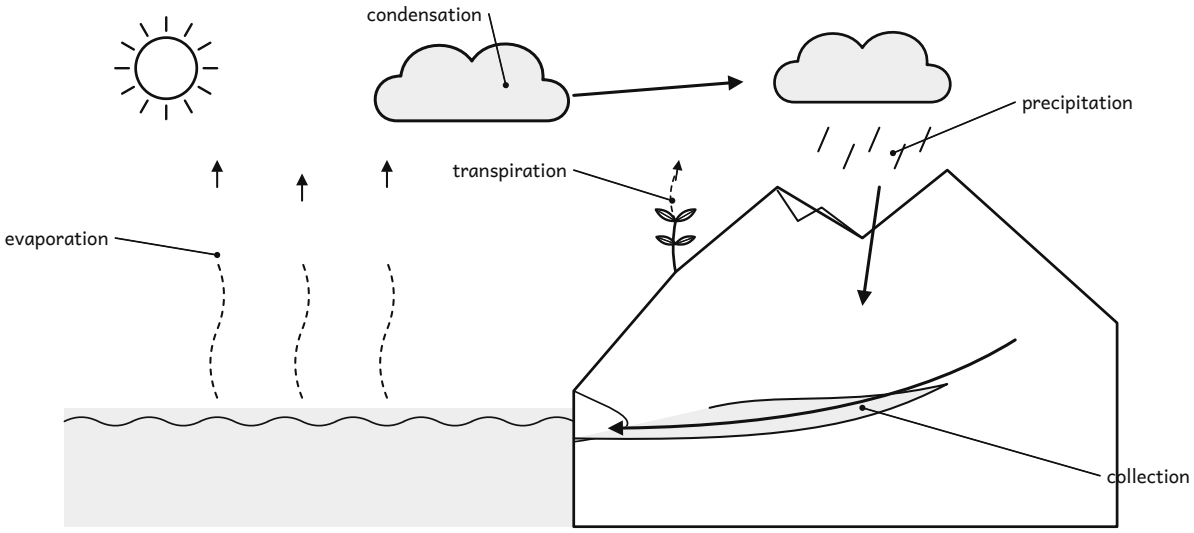
★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Describe the water cycle and the change of state at each step.
- Explain why the water cycle matters to plants, animals and people.
- Discuss how people affect fresh water, and how desalination works.

Bangladesh is a land of water: three great rivers (the Padma, the Jamuna and the Meghna), hundreds of smaller rivers, haors, beels and ponds, and heavy monsoon rain. Yet clean, safe drinking water is still hard to find in many places. Where does water come from, and where does it go?



The water cycle: the same water goes round and round.

Water changes state

Water can be a **solid** (ice), a **liquid** (water) or a **gas** (water vapour). Heating makes ice **melt** at 0 °C and makes water **evaporate**, or **boil** at 100 °C. Cooling makes water vapour **condense** into droplets, and makes water **freeze** at 0 °C.

The water cycle

- **Evaporation:** the Sun heats water in seas, rivers and ponds. Liquid water turns into water vapour (a gas) and rises.
- **Transpiration:** plants take up water through their roots and lose water vapour from their leaves.
- **Condensation:** high in the sky the air is cooler. Water vapour cools and condenses into tiny droplets that make **clouds**.
- **Precipitation:** droplets join together and become heavy, then fall as rain (or hail or snow).
- **Collection:** rain runs over the ground into rivers and the sea, or soaks into the ground as **groundwater**. Then the cycle starts again.

The Sun's energy drives the cycle. The water on Earth today is the same water that dinosaurs drank!

Why the water cycle matters

The water cycle brings **fresh water** to the land. Plants need it to grow, animals need it to drink, and people need it for drinking, cooking, washing, farming and factories. Seas are salty, but evaporation leaves the salt behind, so rain is fresh. Ice in glaciers and snow on mountains store fresh water and release it slowly.

Did you know?

About 97 out of every 100 litres of water on Earth is salty sea water. Most of the rest is frozen. Less than 1 litre in every 100 is fresh water that we can easily use.

Fresh water in Bangladesh

Most of the year's rain falls in the **monsoon** (June to September). Many people drink **groundwater** from tube-wells. Human activities affect both the **quantity** and **quality** of water:

- Pumping too much groundwater in Dhaka makes the water level under the city fall every year.
- Factories, tanneries and sewage have polluted rivers such as the Buriganga.
- In some areas, tube-well water contains **arsenic**, a poison that comes naturally from rocks underground.
- On the coast, rising seas and storm surges push **salt** into ponds, rivers and soil.

Desalination: making salt water drinkable

Desalination removes salt from sea water. One way copies the water cycle: heat the salt water so that water evaporates, then cool the vapour so that it condenses as pure water. The salt is left behind. Another way, called **reverse osmosis**, pushes water through a very fine filter that stops the salt.

Protecting water

We can collect rainwater from roofs, fix dripping taps, never throw rubbish into rivers, and use less water at home. Some coastal villages use **pond sand filters** and rainwater tanks.

Key words

evaporation	liquid changing into a gas
condensation	gas changing into a liquid
precipitation	water falling from clouds as rain, hail or snow
transpiration	plants losing water vapour from their leaves
groundwater	water stored in soil and rocks underground
desalination	removing salt from sea water
water vapour	water in the form of a gas

The water cycle

★☆☆ Grade 5

Name: Class: Date:

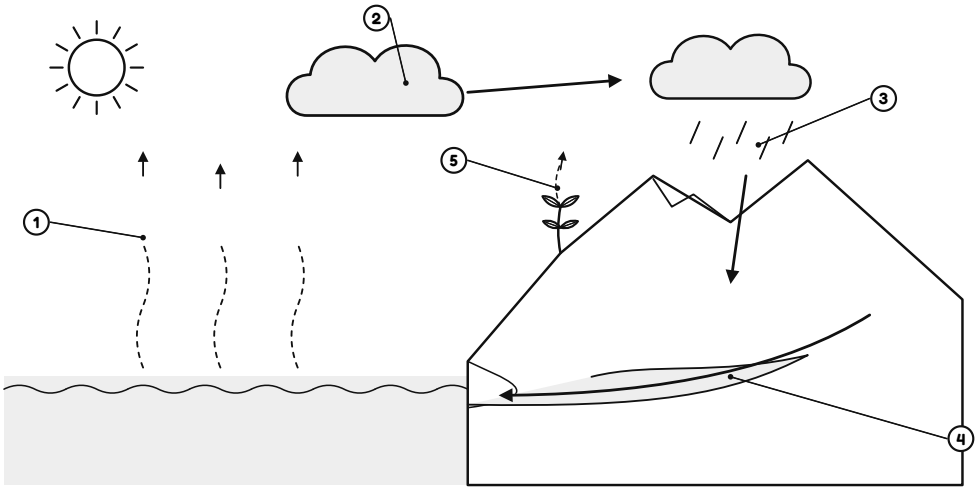
Learning goals

- Label the water cycle.
- Name the changes of state.

1 Label the water cycle. [5]

LABELS

evaporation condensation collection precipitation transpiration



1 _____ 2 _____ 3 _____

4 _____ 5 _____

2 Complete the table. [8]

Change	From	To	Caused by
evaporation			
condensation			
melting			
freezing			

3 Fill in the gaps. [4]

WORD BANK

100 Sun 50 condenses wind 0

Water boils at _____ °C and ice melts at _____ °C. Water vapour cools and _____ to make clouds. The _____ gives the energy that drives the water cycle.

4 **True or false?** [3]

Statement	True	False	If false, correct it
1. Rain water is salty because it comes from the sea.			
2. Plants put water vapour into the air.			
3. Most of Earth's water is fresh water.			

THINK IT THROUGH · B1 · WATER CYCLE

Explaining the cycle

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Order and explain the water cycle.
- Link the cycle to everyday events.

1 **Number the stages to follow one drop of water from the Bay of Bengal back to the sea.** [3]

- ☐ The Sun heats the sea and the water evaporates.
- ☐ The river flows back into the sea.
- ☐ It condenses into droplets and forms clouds.
- ☐ Rain runs into a river.
- ☐ Droplets join and fall as monsoon rain.
- ☐ Water vapour rises and cools.

2 **Explain each observation using the words in brackets.** [6]

a) A cold glass of water gets wet on the outside. (*water vapour, condenses*)

b) Puddles disappear on a sunny day. (*evaporates*)

c) Clothes dry faster on a windy day.

3 **Why is the water cycle important for farmers in Bangladesh?** [2]

4 **How does pumping too much groundwater affect the water cycle in a city? Suggest one solution.** [2]

INVESTIGATE · B1 · WATER CYCLE

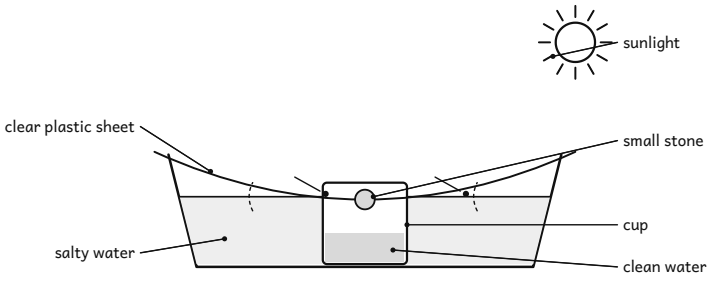
Make a solar still

★★☆ Grade 5

Name: Class: Date:

Learning goals

- Build a model that uses evaporation and condensation.
- Test a variable fairly.
- Use results to explain desalination.



You will need

a large bowl, a cup, salty water, clear plastic sheet, a small stone, sticky tape, a measuring cylinder

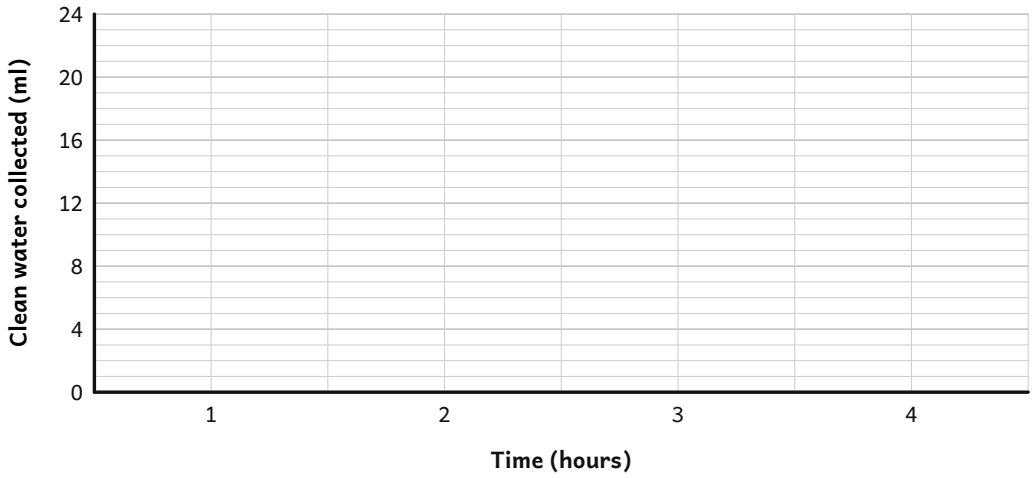
Stay safe

Do not taste anything in a science lab. Test for salt by letting one drop dry on a dark tile.

1 **How does the solar still work? Use the words *evaporates*, *condenses* and *salt*.** [3]

2 **Class 5B set up two identical stills, one in sunshine and one in shade, and measured the clean water every hour.** [4]

Time (hours)	1	2	3	4
In the sun (ml)	4	9	15	20
In the shade (ml)	1	2	4	5



3 **What is the variable they changed? Name two variables they kept the same.** [2]

4 **Write a conclusion and explain it.** [2]

5 **Was the water in the cup salty? How could you check without tasting it?** [1]

REAL-WORLD CASE STUDY · B1 · WATER CYCLE

Too much water, not enough to drink

★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Explain a real water problem using science.
- Compare solutions using evidence.
- Plan actions to protect water.

Science | Geography | Maths | Health | Climate change

In Satkhira and Khulna, in south-west Bangladesh, people are surrounded by water: rivers, shrimp ponds and the sea. But most of it is **too salty to drink**. Storm surges from cyclones such as Aila (2009) flooded ponds with sea water, and rising sea levels push salt further inland each year.

Many women and girls walk several kilometres every day to fetch drinking water. Some families collect **rainwater** from their roofs in big tanks during the monsoon. Villages also use **pond sand filters**, which clean water from a protected pond, and small **desalination plants** that use reverse osmosis.

Drinking salty water for a long time is bad for health, especially for pregnant women, whose blood pressure can rise.

Based on reports about coastal salinity in Bangladesh (WHO, UNICEF, Bangladesh research studies).

1 **Why is there a water problem in a place that has so much water?** [2]

2 **A family of 5 needs about 3 litres of drinking water per person each day.** [3]

a) How much drinking water does the family need in one day?

b) How much in 30 days?

c) Their rainwater tank holds 2000 litres. For about how many days will a full tank last?

3 **Compare three solutions.** [6]

Solution	One advantage	One disadvantage
rainwater tank		
pond sand filter		
desalination plant		

4 **Fetching water often stops girls from going to school. Why is safe water a fairness issue too?** [2]

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

W	N	F	C	D	T	V	A	U	R	M
E	U	F	T	O	P	V	E	E	R	O
R	V	S	N	C	N	N	G	P	A	N
S	B	A	O	O	L	D	M	C	I	S
S	U	B	P	D	V	O	E	J	N	O
S	R	N	C	O	T	A	U	N	G	O
A	E	V	C	U	R	R	P	D	S	N
L	R	G	O	L	H	A	I	O	P	E
T	H	C	W	C	P	S	T	V	U	G
F	M	G	S	H	D	F	S	E	E	R
C	P	P	R	A	A	G	A	K	J	R

cloud **condense** **evaporate** **monsoon** **rain** **river** **salt** **sun** **vapour**

2 **Which process forms clouds?** [1]

- ◯ evaporation
- ◯ condensation
- ◯ melting
- ◯ freezing

3 **What drives the water cycle?** [1]

- ◯ the Moon
- ◯ the wind
- ◯ energy from the Sun
- ◯ the sea

4 **Draw your own water cycle diagram. Label four processes and show the direction with arrows.** [4]

My water cycle

5 **Challenge. Explain why the water cycle is called *a cycle* and why the total amount of water on Earth stays the same.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
describe each step of the water cycle and the change of state	○	○	○
explain dew, drying clothes and clouds	○	○	○
explain why the water cycle matters for life	○	○	○
make a solar still and explain desalination	○	○	○
explain how people affect fresh water in Bangladesh	○	○	○
compare ways to provide safe drinking water	○	○	○